

Akhul Arvind

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Education

University of California, San Diego

Sep 2023 – Jun 2027

B.S. in Neurobiology | B.S. in Cognitive Science [Machine Learning and Neural Computation]

La Jolla, CA

- GPA : 3.9
- Relevant Coursework:
 - **Computational Neurobiology:** neural signal processing & decoding; Poisson/GLM spike-train modeling, tuning curves/population coding, spectral features, and Bayesian state-space decoders.
 - **Neural Signal Processing:** Python course in EEG/LFP/ECOG analysis. Filtering, spectral & time frequency methods (FFT/STFT/Hilbert), coherence/coupling, spike-train analytics for BMI-style systems.

Experience

The Salk Institute - Computational Neurobiology Lab

Dec 2025 – Present

Research Technician

La Jolla, CA

- Designed a disentangled variable autoencoder setup (VAE: $z_c z_{style}=12, z_{batch}=4$) with gene ontology-specific masking to identify treatment-caused biological signals using latent node perturbation.
- Created a developmental slope divergence analysis pipeline that ranks genes that are most perturbed across various ages. Validated top 50 hits in each time point against 2 independently published datasets.
- Recording from in-vitro mouse brain slices using multi-electrode arrays; constructing activity analysis pipelines with temporal convolutional networks & Bayesian inference over electrode dense time series.

De Sa UCSD BCI Lab

Feb 2026 – Present

Research Assistant

La Jolla, CA

- Producing reconstructed viewed images by regressing time-series neural signal from a 128-channel EEG into 768-dim CLIP-Vision and 36,864 VAE latent space nodes using BayesianRidge correction ($K=50$).
- Created a hierarchical Bayesian reconstruction component to compare the previous standard regression baseline to probabilistic reconstruction to generalize past each participant's neural response signatures.

Swartz Center for Computational Neuroscience (SCCN)

Feb 2026 – Present

VR EEG Lab Research Assistant

La Jolla, CA

- Refine signal processing and classification pipelines for live brain-state discernment for individual participants playing a VR puzzle game; Common spatial patterns, sliding window segmentation, ICA Artifact rejection, spectral feature extraction, and event-based analysis using 64-channel EEG cap.
- Extract classified neural states to deploy an adaptive AI agent that can help the participant complete tasks; detects stress in real time and performs task performance interventions.

Projects

MRI Brain Lesion Segmentation and Detection

May 2025 – Aug 2025

- Built a cross-site MRI pipeline in Python (NiBabel, SimpleITK) that converts raw DICOM to BIDS-organized NIfTI with N4 bias correction, skull stripping, MNI alignment, and intensity normalization — designed to generalize across hospital scanners.
- Train 3D U-Net and 3D ResNet models in PyTorch/MONAI with patch-based sampling and targeted augmentations for lesion segmentation and classification. Evaluate on Dice, AUC, and calibration metrics.
- Validate on fully held-out hospital sites to test cross-site robustness. Version all code and data, track experiments end-to-end for reproducibility.

Technical Skills

- **ML / Deep Learning:** Python | PyTorch | MONAI | scikit-learn | BayesianRidge | VAE / disentangled VAE | temporal CNNs | ICA | spectral feature extraction | time-frequency analysis | patch-based 3D segmentation | NumPy | SciPy | pandas | MLflow | Docker
- **Neuro / Data:** EEG time-series decoding | multi-electrode array recordings | MRI preprocessing (NiBabel, SimpleITK, ANTs, FSL, FreeSurfer) | BIDS/NIfTI/DICOM | single-cell RNA-seq (scanpy) | EEGLAB | MATLAB
- **Engineering / Tools:** Git | Slurm/HPC (DSMLP, SDSC Expanse) | FastAPI | React | Three.js | Next.js | Supabase | TypeScript | Linux/Bash